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dda5796 · 9 minutes ago



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# Assignment 4: Prompt Engineering Experiments

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**Internship:** AI Engineering Fellowship 2026

**Organization:** Invictus Solutions

## Introduction

Prompt engineering is the process of designing effective prompts that help Large Language Models (LLMs) generate accurate, useful, and well-structured responses. This report demonstrates five different prompt engineering techniques.

## Experiment 1 – Role Prompting

## Prompt

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You are an experienced AI Engineering mentor.



Explain Artificial Intelligence to a beginner Software Engineering student using simple language and real-life examples.

## Output

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Artificial Intelligence (AI) enables computers to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, and language understanding. Examples include ChatGPT, self-driving cars, recommendation systems, and virtual assistants.

## Observation

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- The AI responded as an expert mentor.
- The explanation was easy to understand.
- Real-world examples made the concept clearer.

## Improvements

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- Specify the education level of the audience.
- Request diagrams or bullet points for better understanding.

# Experiment 2 – Chain of Thought Prompting

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## Prompt

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Solve this problem step by step.



A company has 120 employees.

35% work remotely.

How many employees work remotely?

## Output

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Step 1: Calculate 35% of 120.

Step 2:  $120 \times 35 \div 100 = 42$ .

Final Answer: 42 employees work remotely.

## Observation

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- The AI explained every calculation.
- The reasoning was easy to follow.
- The final answer was accurate.

## Improvements

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- Ask the AI to verify the final answer.
- Request additional explanation if teaching beginners.

# Experiment 3 – Few-Shot Prompting

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## Prompt

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Example 1:  
Input: Hello  
Output: Greeting

Example 2:  
Input: Bye  
Output: Farewell

Now classify:

Input: Good Morning



## Output

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Greeting

## Observation

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- The examples helped the AI understand the task.
- The classification was correct.

## Improvements

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- Add more examples.
- Include different categories for better accuracy.

# Experiment 4 – Structured Output (JSON)

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## Prompt

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Generate student information in JSON format.



Name: Hassan Murtaza Zaidi

University: The University of Faisalabad

Department: Software Engineering

Semester: 4

## Output

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```
{  
  "name": "Hassan Murtaza Zaidi",  
  "university": "The University of Faisalabad",  
  "department": "Software Engineering",  
  "semester": 4  
}
```



## Observation

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- The output followed proper JSON structure.
- JSON format is useful for APIs and software applications.

## Improvements

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- Validate the JSON before using it.
- Add more fields if required.

## Experiment 5 – Prompt Optimization

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### Original Prompt

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Tell me about AI.



### Output

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A general explanation of Artificial Intelligence.

### Optimized Prompt

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You are an AI professor.



Explain Artificial Intelligence to a second-year Software Engineering student.

Include:

- Definition
- Types of AI
- Real-world applications
- Advantages
- Challenges

Limit the answer to 300 words and use headings and bullet points.

### Output

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The AI produced a structured explanation with headings, bullet points, and detailed information.

## Observation

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- The optimized prompt produced a much better response.
- The output was more organized and informative.

## Improvements

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- Clearly define the audience.
- Specify the response format.
- Add word limits and formatting instructions.

## Conclusion

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This assignment demonstrated five important prompt engineering techniques:

- Role Prompting
- Chain of Thought Prompting
- Few-Shot Prompting
- Structured Output (JSON)
- Prompt Optimization

These techniques improve the quality, consistency, and usefulness of AI-generated responses.